

CityTech TTP Summer Bootcamp

Visualizing Data with seaborn

2026-07-13

Agenda

1. Review
2. Why Visualize?
3. Meet seaborn
4. A Tour of seaborn Plots
5. Customizing Plots
6. Code Together and Pair Project

Review

What are the most important things you remember from last week?

Why Visualize?

We can print data as a **table** or crunch it into **summary statistics**. So why bother with charts?

Let's look at one year of bike-share rides **three ways**, and each time ask:

What is the story in this data?

Every number in this example is invented for teaching, not real data.

The Raw Data

A year of daily bike-share rides. Every value is here, but what is the story?

date	day	rides	members	casual	avg_min	high_F
2025-01-01	Wed	8,214	6,102	2,112	10.8	41
2025-01-02	Thu	11,905	9,431	2,474	11.3	39
2025-01-03	Fri	12,338	9,880	2,458	12.0	44
2025-01-04	Sat	9,672	5,901	3,771	13.5	46
2025-01-05	Sun	8,431	5,204	3,227	12.9	43
...						(365 rows)

(Invented data, not real.)

The Summary

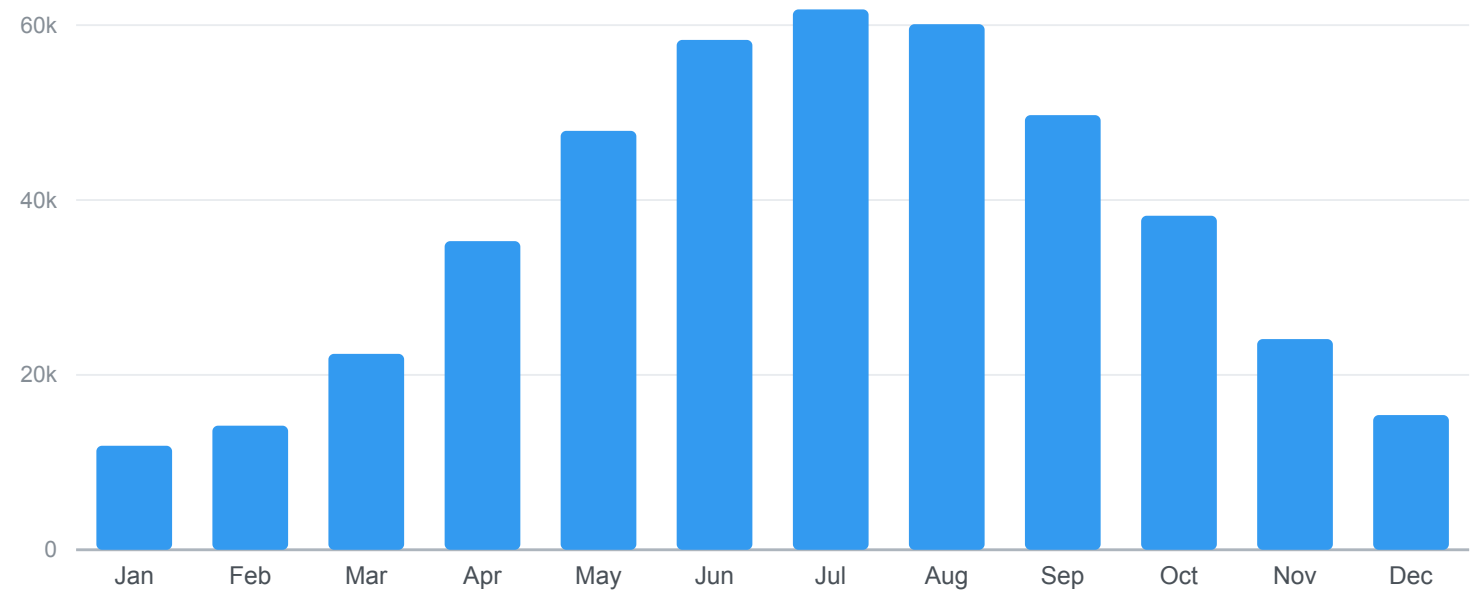
Aggregated to one number per month, sorted by average daily rides.

Month	Avg rides
Jul	61,800
Aug	60,100
Jun	58,300
Sep	49,700
May	47,900
Oct	38,200

Month	Avg rides
Apr	35,300
Nov	24,100
Mar	22,400
Dec	15,400
Feb	14,200
Jan	11,900

(Invented data, not real.)

The Same Data, Visualized



(Invented data, not real.)

Why It Matters

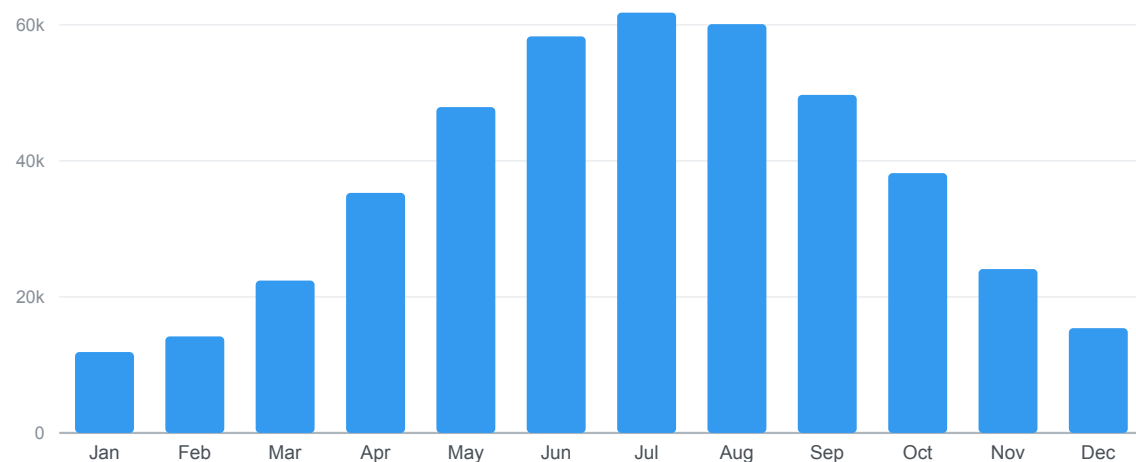
Same data, three ways:

- The **raw table** holds everything, but hides the story
- The **summary** is compact, but you still have to read and interpret it
- The **chart** shows it in one second: rides **peak in summer**, bottom out in winter

Visualization is how we see structure: trends, cycles, and outliers that raw numbers bury.

Aside: Data Informs Future Decisions

Data science is not only about describing the past; it is about **deciding what to do next**.



Discuss: how could you use this data to make better decisions for next year?

(Invented data, not real.)

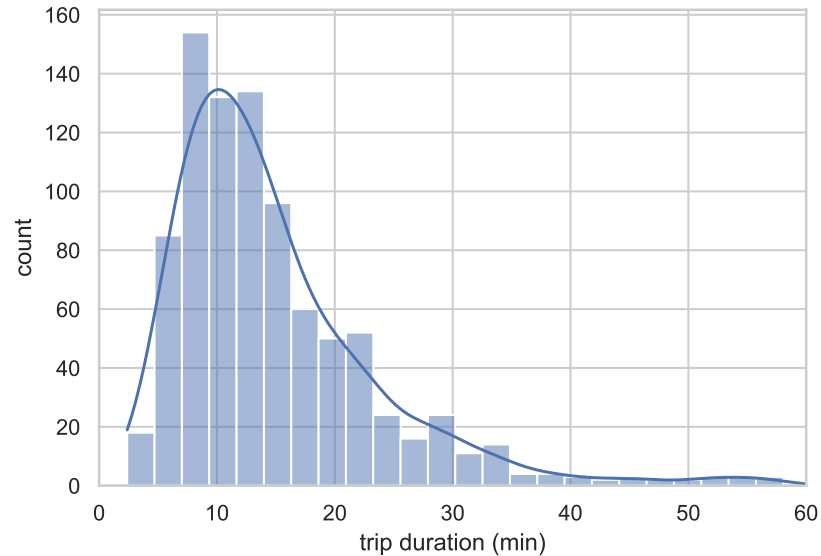
Meet seaborn

seaborn is a Python library for **statistical data visualization**, built on top of matplotlib.

- Works **directly with pandas** DataFrames: name your columns, it handles the rest
- **High-level**: good-looking, sensible charts in a line or two
- Built for **exploratory data analysis**: distributions, comparisons, relationships
- Smart defaults for colors, labels, and styling, so you focus on the data

```
import seaborn as sns  
  
sns.barplot(data=df, x="month", y="rides")    # the chart we just saw
```

Histogram



(Invented data, not real.)

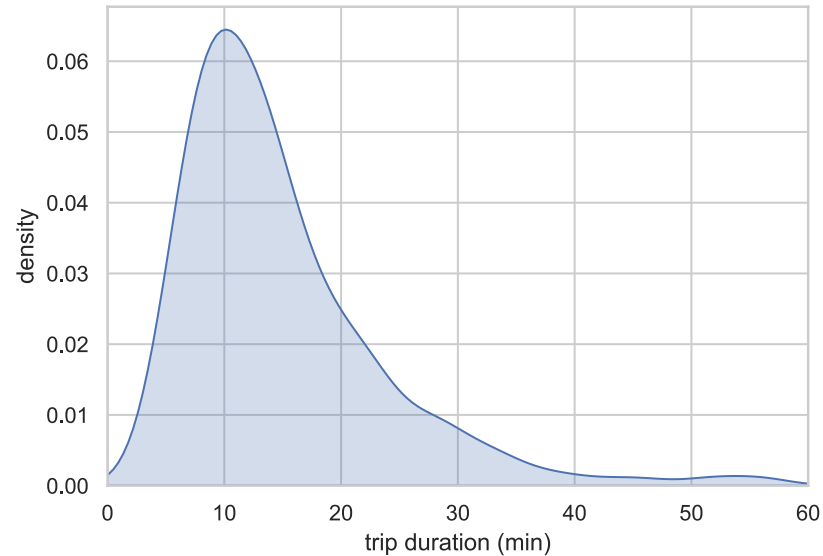
Best for: the distribution of one numeric variable

Questions it answers

- Where do most values fall?
- Is the data skewed or symmetric?
- Are there outliers?

```
sns.histplot(data=df, x="duration", kde=True)
```

Density (KDE)



(Invented data, not real.)

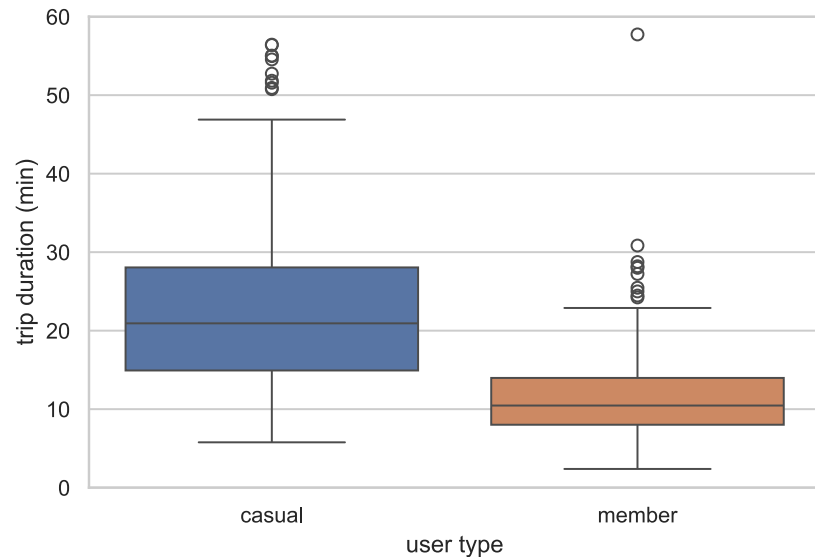
Best for: a smooth view of one variable's distribution

Questions it answers

- What is the overall shape?
- Where is the data most concentrated?
- Is there one peak or several?

```
sns.kdeplot(data=df, x="duration", fill=True)
```

Box Plot



Best for: comparing distributions across groups, compactly

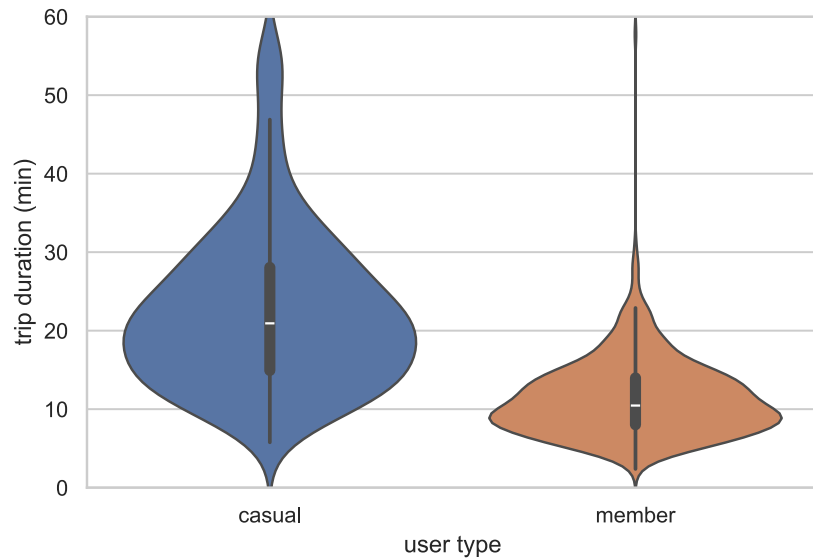
Questions it answers

- How do the medians compare?
- How spread out is each group?
- Which points are outliers?

```
sns.boxplot(data=df, x="user_type", y="duration")
```

(Invented data, not real.)

Violin Plot



(Invented data, not real.)

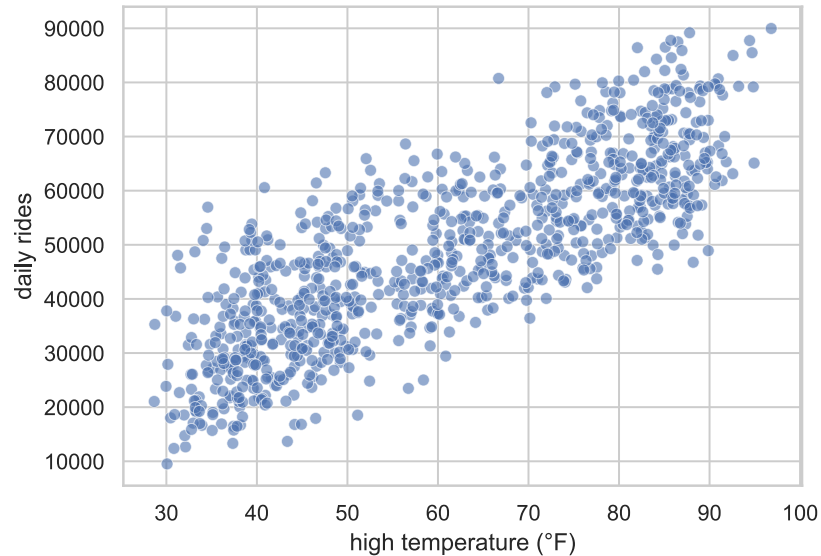
Best for: comparing the full distribution shape across groups

Questions it answers

- How does each group's shape differ?
- Is a group bunched or spread out?
- Is any group bimodal?

```
sns.violinplot(data=df, x="user_type", y="duration")
```

Scatter Plot



(Invented data, not real.)

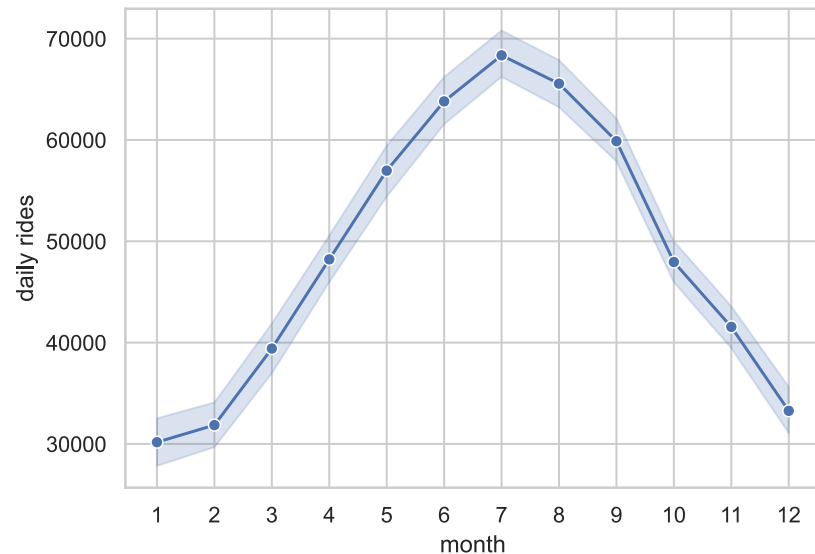
Best for: the relationship between two numeric variables

Questions it answers

- Do the two move together?
- Is the relationship positive or negative?
- Are there clusters or outliers?

```
sns.scatterplot(data=df, x="temp", y="rides")
```

Line Plot



(Invented data, not real.)

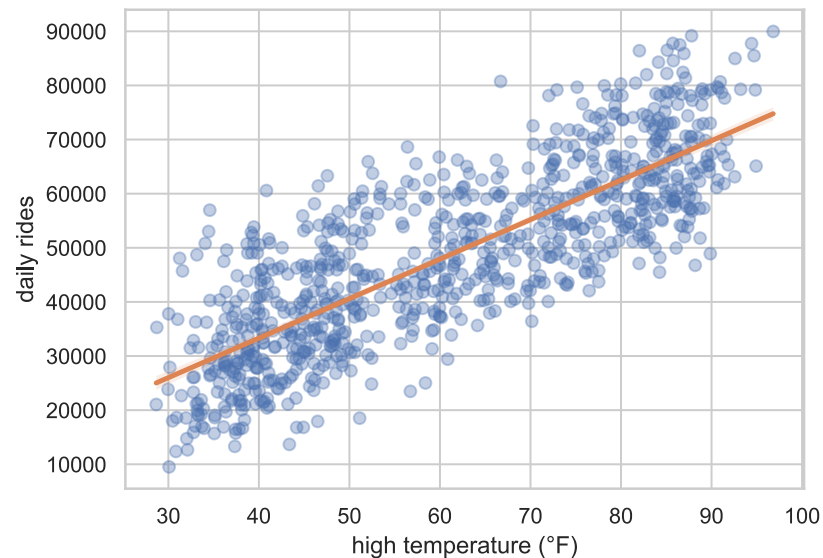
Best for: how a value changes over an ordered axis, like time

Questions it answers

- Is it trending up or down?
- Is there a repeating cycle?
- When did it peak or dip?

```
sns.lineplot(data=df, x="month", y="rides")
```


Regression Plot



(Invented data, not real.)

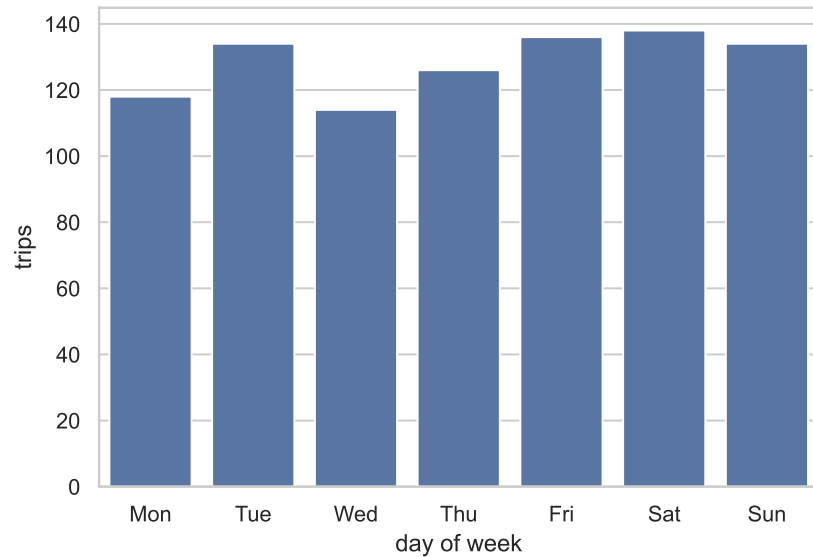
Best for: a relationship plus a fitted trend line

Questions it answers

- What is the trend, and how steep?
- How strong is the relationship?
- What is Y at a given X?

```
sns.regplot(data=df, x="temp", y="rides")
```

Count Plot



(Invented data, not real.)

Best for: how many rows fall in each category

Questions it answers

- Which category is most common?
- Which is rare?
- Is the data balanced across categories?

```
sns.countplot(data=df, x="day")
```

Heatmap



(Invented data, not real.)

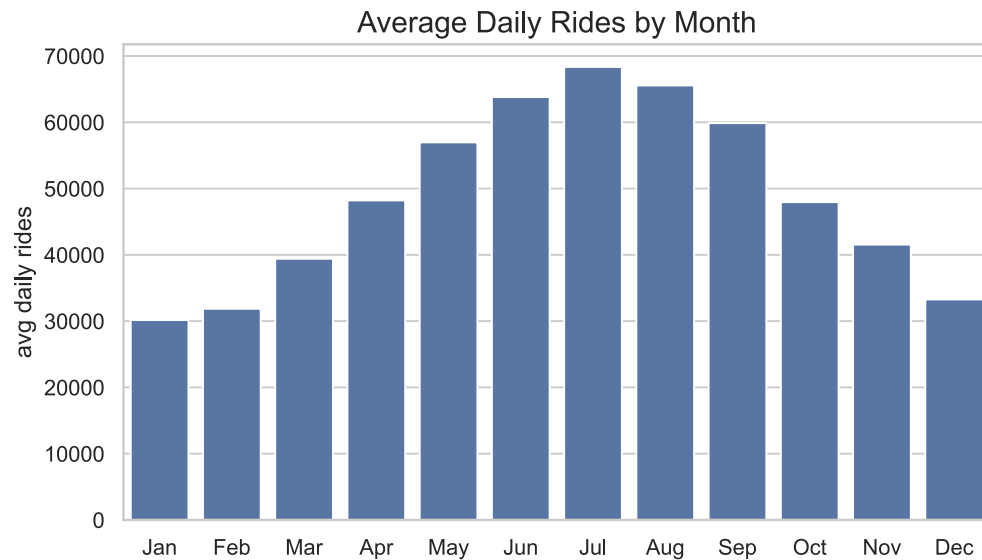
Best for: a grid of values shown as color, like a correlation matrix

Questions it answers

- Which variables move together?
- Where are the strong or weak links?
- Any surprising hot or cold spots?

```
sns.heatmap(df.corr(), annot=True)
```

Customizing: A Few Changes



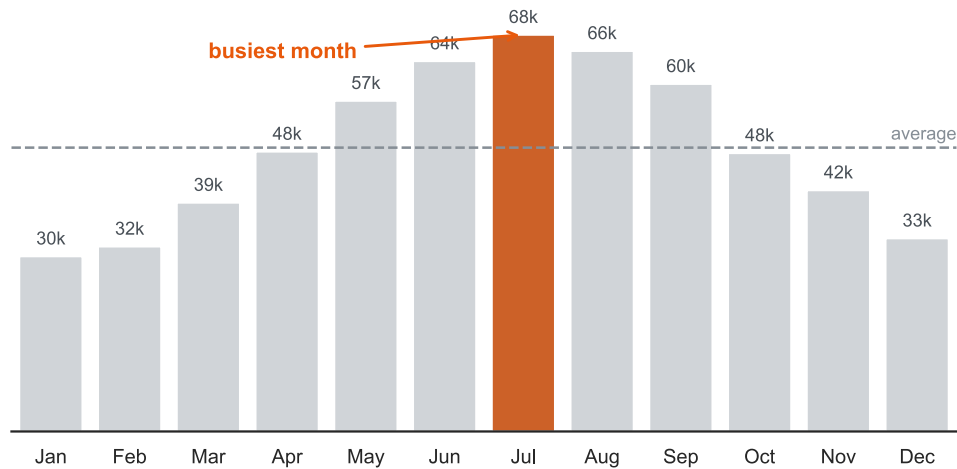
(Invented data, not real.)

A few lines on top of **one seaborn call**:

```
sns.set_theme(style="whitegrid")
ax = sns.barplot(data=m, x="month",
                  y="rides", color="#4c72b0")
ax.set_title("Average Daily Rides by Month")
ax.set_ylabel("avg daily rides")
ax.set_xlabel("")
```

Customizing: Going Further

When New Yorkers Ride



(Invented data, not real.)

The same `barplot`, but seaborn hands you a `matplotlib Axes` to restyle everything:

- muted bars, the **peak** highlighted
- **value labels** on each bar
- an **average** reference line
- an **annotation** on the busiest month

Aside: Built on matplotlib

seaborn does not draw anything itself. It is a friendly layer on top of **matplotlib**.

- **matplotlib** is Python's foundational plotting library: powerful, but low-level and verbose
- seaborn calls it for you, with **smart defaults** and far **less code**
- Every seaborn chart is really a matplotlib **Figure** and **Axes**
- So you can drop down to matplotlib to **fine-tune anything** (as we just did)

seaborn for speed, matplotlib for total control.

Learn More: Official Docs

- **seaborn:** seaborn.pydata.org
- **matplotlib:** matplotlib.org

Both sites have an **example gallery**: browse the pictures, find the plot you want, and copy its code as a starting point.

Code Together: NYC Open Data

Let's take a **real** dataset from raw to insight together, live.

- **Find:** browse [NYC Open Data](#) and pick a dataset
- **Explore:** `head()`, `info()`, `describe()`, `value_counts()`
- **Clean:** fix types, missing values, duplicates, outliers
- **Visualize:** chart it with **seaborn** to surface the story
- **Insight:** ask a question and let the data answer

Pair Programming: Data for a Better NYC

Pair up, take a real dataset from raw to insight, then **present to the class**. Use data to propose a change that would make life better for NYC residents.

Your presentation should cover:

1. Your **initial questions**
2. How you **chose your dataset**
3. What you did while **cleaning**
4. Your **final visualizations**
5. How your **question evolved** (or didn't)
6. Your **final recommendations**

Your Notebook Is Your Report

Build your notebook to be **read top to bottom**, like a report: interleave narration with your data and charts.

- **Markdown cells** for the story: headings, your questions, what you found
- **Code cells** for the work: the DataFrame, the cleaning, the seaborn plots
- Alternate them so a reader can follow your reasoning

Make a cell Markdown: select it, press `Esc` then `M` (or use the cell-type dropdown), then `Shift+Enter` to render.

Docs: [Working With Markdown Cells](#)

Today's Exit Ticket

- ☐ Explain why a chart can reveal patterns a table or summary hides
- ☐ Create a **seaborn** plot from a pandas DataFrame
- ☐ Match a question to the right plot (distribution, relationship, comparison)
- ☐ Customize a plot: title, labels, colors, and annotations
- ☐ Use a real dataset to propose and present a data-backed recommendation